CS 70 离散数学 和 概率论
笔记 16
春季 2026 Course Notes
Random Variables: 分布 和 期望
Recall our setup of a probabilistic experiment as a procedure of drawing a sample from a 集合 of possible values, andassigning aprobabilityforeachpossible outcomeoftheexperiment. Forexample, 如果 wetossa

faircoinntimes,thenthereare2npossibleoutcomes,eachofwhichisequallylikelyandhasprobability
1.
2n
Now 假设 we want to make a measurement in our experiment. 对于 例子, we can ask what is the number of heads in n coin tosses; call this number X. Of course, X is 非 a fixed number, 但是 it depends on the actual 序列 of coin flips that 我们得到. 对于 例子, 如果 n = 4 和 we observe the 结果
ω =HTHH, 那么 X =3; whereas 如果 we observe the 结果 ω =HTHT, 那么 X =2. In this 例子 of n coin tosses, we only know that X is an 整数 between 0 和 n, 但是 we do 非 know what its exact value is 直到 we observe 其中 结果 of n coin flips is realized 和 count how many heads 存在.
Because every possible 结果 is assigned a 概率, the value X also carries with it a 概率 对于 each possible value it can take. The 表 below lists all the possible values X can take in the 例子 of n=4cointosses,alongwiththeirrespectiveprobabilities.

outcomesω	valueofX (#heads)	probabilityofoccurring
TTTT	0	1/16
HTTT, THTT, TTHT, TTTH	1	4/16
HHTT, HTHT, HTTH, THHT, THTH, TTHH	2	6/16
HHHT, HHTH, HTHH, THHH	3	4/16
HHHH	4	1/16

Such a value X that depends on the 结果 of the probabilistic experiment is called a random 变量 (abbreviated r.v.). As we see from the 例子 above, a random 变量 X typically does 非 have a definitive value, 但是 instead only has a 概率 分布 over the 集合 of possible values X can take,
其中 is why it is called random. 所以 the 问题 “What is the number of heads in n coin tosses?” does 非 exactly make sense because the 答案 X is a random 变量. 但是 the 问题 “What is the typical numberofheadsinncointosses?” makessense—itisaskingwhatistheaveragevalueofX (thenumber of heads) 如果 we repeat the experiment of tossing n coins multiple times. This average value is called the

expectationofX,andisoneofthemostusefulsummaries(alsocalledstatistics)ofanexperiment.

1 Random Variables

Before we formalize the above notions, 令 us 考虑 another 例子 to enforce our conceptual under-standingofarandomvariable.

例子: FixedPointsofPermutations

问题:
Supposewecollectthehomeworksofnstudents,randomlyshufflethem,andreturnthemtothe

students. Howmanystudentsreceivetheirownhomework?

Heretheprobabilityspaceconsistsofalln!permutationsofthehomeworks,eachwithequalprobability
1.
n!

Ifwelabelthehomeworksas1,2,...,n,theneachsamplepointisapermutation π
=(π ,..., π)where π
1 n i
is the 作业 即 returned to the ith student. We call i a fixed
point of π 如果 π =i, i.e., 如果 student i
i
receivestheirownhomework.
As in the coin flipping case above, our 问题 does 非 have a
simple numerical 答案 (such as 4),

becausethenumberdependsontheparticularpermutationwechoose(i.e.,onthesamplepoint).
Letuscall
thenumberoffixedpointsX
,whichisarandomvariabletakingvaluesintheset{0,1,2,...,n}.
(Actually
n
thevalueX =n 1isnotpossible: why?)
n
FormalDefinitionofaRandomVariable
Wenowformalizetheconceptsdiscussedabove.
定义 16.1 (Random 变量). A randomvariable X on a 样本空间 is a 函数
X: →R
thatassignstoeachsamplepoint ω ∈ arealnumberX(ω).

Untilfurthernotice,wewillrestrictourattentiontorandomvariablesthatarediscrete,i.e.,theytakevalues
inarangethatisfiniteorcountablyinfinite.
ThismeanseventhoughwedefineX tomap toR,theactual
setofvalues{X(ω): ω ∈ }thatX takesisadiscretesubsetofR.
A random 变量 can be visualized in general by the picture in 图
1.1 笔记 that the term “random
变量” is really something of a misnomer: it is a 函数 所以 存在
nothing random about it 和 it is
definitelynotavariable!
Whatisrandomiswhichsamplepointoftheexperimentisrealizedandhencethe
valuethattherandomvariablemapsthesamplepointto.
Randomvariable
X
Sample
Space R
a

X =#offixedpoints
3
123 231
R
132 312
0 1 2 3
213 321
Samplespaceof
allpermutations
Figure1:
Visualizationofhowarandomvariableisdefinedonthesamplespace.

练习. By completing the lower picture in 图 1, 证明 that the only
possible values 对于 the r.v. X
3
are0,1and3,andthattheirprobabilitiesare 1, 1 和
1,respectively.
3 2 6

1Thefiguresinthisnoteareinspiredbyfiguresinchapter2ofIntroductiontoProbabilitybyD.BertsekasandJ.Tsitsiklis.

2 概率 分布

Whenweintroducedthebasicprobabilityspaceinanearliernote,wedefinedtwothings:
1.
Thesamplespace consistingofallthepossibleoutcomes(samplepoints)oftheexperiment.
2. Theprobabilityofeachofthesamplepoints.

Analogously,therearetwoimportantthingsaboutanyrandomvariable:
1. Thesetofvaluesthatitcantake.
2. Theprobabilitieswithwhichittakesonthevalues.
因为 a random 变量 is defined on a 概率 space, we can calculate
these probabilities 给定 the
probabilitiesofthesamplepoints.
LetabeanynumberintherangeofarandomvariableX. Thentheset
{ω ∈ :X(ω)=a}
is an 事件 in the 样本空间 (because it is a 子集 of). We usually
abbreviate this 事件 to simply
“X = a”. 因为 X = a is an 事件, we can talk about its 概率, P[X =
a]. The collection of these

probabilities,forallpossiblevaluesofa,isknownasthedistributionoftherandomvariableX.
定义 16.2 (分布). The 分布 of a 离散的 random 变量 X is the collection
of values
{(a,P[X =a]):a∈A},whereA
isthesetofallpossiblevaluestakenbyX.
因此,thedistributionoftherandomvariableX
inourpermutationexampleaboveis:
1 1 1
P[X =0]= ; P[X =1]= ; P[X =3]= .
3 2 6
Ifneeded,wemayalsowriteP[X =a]=0forallothervaluesofa.
The 分布 of a random 变量 can be visualized as a bar diagram,
shown in 图 2. The x-axis
representsthevaluesthattherandomvariablecanassume.
Theheightofthebaratavalueaistheprobability
P[X =a]. Each of these probabilities can be computed by
looking at the 概率 of the corresponding
eventinthesamplespace.
NotethatthecollectionofeventsX
=a,fora∈A,satisfytwoimportantproperties:
• AnytwoeventsX =a andX =a witha =a aredisjoint.
1 2 1 2
• Theunionofalltheseeventsisequaltotheentiresamplespace .
The collection of events 因此 form a 划分 of the 样本空间 (see 图 2).
Both properties follow
directly from the fact that X is a 函数 defined on , i.e., X
assigns a unique value to each 和 every
possible sample point in . 所以, when we sum up the
probabilities of the events X = a, we are really

summinguptheprobabilitiesofallthesamplepoints,givingusatotalofexactly1.

P[X =a]
X =a
1
X =a
2
X =a
3
a a a
1 2 3
P[X =a]
1
123 231 2
1
132 312 3
1
6
213 321
0 1 2 3
Figure2:
Visualizationofhowthedistributionofarandomvariableisdefined.
BernoulliDistribution
A simple yet very useful 概率 分布 is the Bernoulli 分布 of a
random 变量 其中
takesvaluein{0,1}:
(cid:40)
p, ifi=1,
P[X =i]=
1 p, ifi=0,
where0≤ p≤1. WesaythatX
isdistributedasaBernoullirandomvariablewithparameter
p,andwrite
X   Bernoulli(p) 或 X   Ber(p).
BinomialDistribution

Letusreturntoourcointossingexampleabove,wherewedefinedourrandomvariableX
tobethenumber
of heads. More formally, 考虑 the random experiment consisting
of n 独立的 tosses of a biased
cointhatshowsH withprobability p. Eachsamplepointω
isasequenceoftosses,andX(ω)isdefinedto
bethenumberofheadsinω. Forexample,whenn=3,X(THH)=2.
To 计算 the 分布 of X, we first enumerate the possible values
that X can take. They are simply
0,1,...,n. 那么 we 计算 the 概率 of each 事件 X =i 对于 i=0,1,...,n.
The 概率 of the
事件 X =i is the sum of the probabilities of all the sample
points with exactly i heads (对于 例子, 如果
n=3andi=2,therewouldbethreesuchsamplepoints{HHT,HTH,THH}).
Anysuchsamplepointhas
概率 pi(1 p)n i,sincethecoinflipsareindependent.
Thereareexactly (cid:0)n(cid:1) ofthesesamplepoints.
i
因此,
(cid:18) (cid:19)
n
P[X =i]= pi(1 p)n i, fori=0,1,...,n. (1)
i

This 分布, called the 二项的 分布, is one of the most important
distributions in 概率.

Arandomvariablewiththisdistributioniscalledabinomialrandomvariable,andwewrite
X Bin(n,p),
wherendenotesthenumberoftrialsand
ptheprobabilityofsuccess(observinganH intheexample). An
例子 of a 二项的 分布 is shown in 图 3. Notice that due to the
properties of X mentioned
above,itmustbethecasethat∑ n P[X =i]=1,whichimpliesthat∑ n
(cid:0)n(cid:1) pi(1 p)n i=1. Thisprovides
i=0 i=0 i
a probabilistic 证明 of the 二项的 定理 from an earlier 笔记 其中 we
saw it combinatorially, 对于
a= pandb=1 p.
P[X =a] P[X =a]
Binomialdistribution Binomialdistribution
n=9, p= 1 nlarge, psmall
2
a a
1 2 3 4 5 6 7 8 9 n
Figure3: Thebinomialdistributionsfortwochoicesof(n,p).

Althoughwedefinethebinomialdistributionintermsofanexperimentinvolvingtossingcoins,
thisdistri-
butionisusefulformodelingmanyreal-worldproblems.
Considerforexampletheerrorcorrectionproblem
studied earlier. 回忆 we wanted to encode n packets into n+k
packets 满足 the recipient can
reconstructtheoriginalnpacketsfromanynpacketsreceived.
Butinpractice,thenumberofpacketlosses
is random, 所以 how do we choose k, the amount of redundancy?
如果 we model each packet getting lost with
概率 p 和 the losses are 独立的, 那么 如果 we transmit n+k packets,
the number of packets re-
ceivedisarandomvariableX withbinomialdistribution: X
Bin(n+k,1 p)(wearetossingacoinn+k
times, 和 each coin turns out to be a head (packet received)
with 概率 1 p). 所以 the 概率 of
successfullydecodingtheoriginaldatais:
n+k n+k(cid:18) n+k (cid:19)
P[X ⩾n]= ∑P[X =i]= ∑ (1 p)ipn+k i.
i
i=n i=n
Givenfixednand
p,wecanchoosekesuchthatthisprobabilityisnolessthan,say,0.99.
HypergeometricDistribution
考虑 an urn containing N = B+W balls, 其中 B balls are black 和 W
are white. 假设 you
randomly sample n⩽N balls from the urn with replacement, 和 令
X 表示 the number of black balls
in your sample. What is the 概率 分布 of X? 因为 the 概率 of seeing
a black ball is

B/N 对于 each draw, independently of all other draws, X
follows the 二项的 分布 Bin(n,p), 其中
p=B/N.
What 如果 you randomly sample n⩽N balls from the urn without
replacement? In this case, the 概率
of seeing a black ball in the ith draw depends on the colors

inthecaseofsamplingwithreplacement,thedrawsarenotindependent.
Theprobabilitydistributionofthe
numberY ofblackballsinthissettingcanbefoundasfollows.
Considerasequenceω
ofndrawswherethefirstkballsareblackandthenextn kballsarewhite.
那么,

usingtheProductRulefromNote14,theprobabilityofthisparticularsequenceofdrawscanbecomputed
as
(cid:0)B(cid:1)(cid:0)W (cid:1)
B B 1 B k+1 W W 1 W (n k)+1 1
P[ω]= × ×···× × × ×···× = k n k . (2)
N N 1 N k+1 N k N k 1 N n+1
(cid:0)N(cid:1) (cid:0)n(cid:1)
n k
此外, anyothersequenceω′ consistingofk blacksballsandn k
whiteballshasexactlythesame
probabilityasω:
ifwecarryoutasimilarcalculationasin(2)forthesequenceofdrawsinω′,thenumerators

appearinginP[ω′]willbesomepermutationofthenumeratorsinthefirstlineof(2),whilethedenominators
will be exactly the same as in (2). 因为 存在
(cid:0)n(cid:1)
distinct sequences of length n consisting of k black
k
ballsandn kwhiteballs,weobtain
(cid:18) (cid:19) (cid:0)B(cid:1)(cid:0)N B(cid:1)
n
P[Y =k]= P[ω]= k n k , (3)
k
(cid:0)N(cid:1)
n
fork∈{0,1,...,n}. (Notethat (cid:0)m(cid:1) =0if j>m,soP[Y
=k] =0onlyifmax(0,n+B N)≤k≤min(n,B).)
j

ThisprobabilitydistributioniscalledthehypergeometricdistributionwithparametersN,B,n,andwewrite
Y Hypergeometric(N,B,n).
3 Multiple Random Variables, Independence, 和 Exchangeability
Often one is interested in multiple random variables on the
same 样本空间. 考虑, 对于 例子,
the 样本空间 of flipping three coins. One could 定义 many random
variables: 对于 例子 a random
variableX indicatingthenumberofheads,orarandomvariableY
indicatingthenumberoftails,orabinary
randomvariableZ indicatingwhetherthefirsttossisH ornot.
Notethatforeachsamplepoint,anyrandom
variablehasaspecificvalue: e.g.,forω
=HTT,wehaveX(ω)=1,Y(ω)=2,andZ(ω)=1.

Theconceptofadistributioncanthenbeextendedtoprobabilitiesforthecombinationofvaluesformultiple
randomvariables.
Definition16.3.
ThejointdistributionoftwodiscreterandomvariablesX andY
isthecollectionofvalues
{((a,b),P[X =a,Y =b]):a∈A, b∈B}, 其中 A is the 集合 of all
possible values taken by X 和 B is

thesetofallpossiblevaluestakenbyY.
GivenajointdistributionofX andY,thedistributionP[X =a]ofX
iscalledthemarginaldistributionofX,
andcanbefoundbysummingoverthevaluesofY. Thatis,
P[X =a]= ∑ P[X =a,Y =b].
b∈B
ThemarginaldistributionofY isdefinedanalogously.

AjointdistributionoveranysetofrandomvariablesX ,...,X
(forexample,X couldbethevalueoftheith
1 n i
rollofasequenceofndierolls)isP[X =a ,...,X =a],wherea ∈A
andA isthesetofpossiblevalues
1 1 n n i i i
forX. ThemarginaldistributionofX
canbeobtainedbysummingoverallthepossiblevaluesoftheother
i i
variables.
Independence

Independenceforrandomvariablesisdefinedinananalogousfashiontoindependenceforevents:
定义 16.4 (Independence). Random variables X 和 Y on the same
概率 space are said to
be 独立的 如果 the events X =a 和 Y =b are 独立的 对所有 values a,b.
Equivalently, the joint
distributionofindependentr.v.’sdecomposesas
P[X =a,Y =b]=P[X =a]P[Y =b], a,b.
Mutualindependenceofmorethantwor.v.’sisdefinedsimilarly.

Averyimportantexampleofindependentrandomvariablesareindicatorrandomvariablesforindependent
events. 如果 I denotes the indicator r.v. 对于 the ith toss of a
coin being H, 那么 I ,...,I are mutually inde-
i 1 n
pendentrandomvariables.
Thisexamplemotivatesthecommonlyusedphrase“independentandidentically
distributed (i.i.d.) 集合 of random variables.” In this 例子, {I
,...,I } is a 集合 of i.i.d. indicator random
1 n
variables.
Exchangeability
假设 we randomly sample without replacement n ⩽ N balls from
an urn containing B black balls
和 N B white balls. We showed earlier that the number Y of
black balls in the sample follows the
Hypergeometric(N,B,n)分布. Notethatwecanwrite
Y =I +• • • +I ,
1 n
whereI
istheindicatorrandomvariablethatequals1iftheithdrawisblackand0otherwise.
i
Bytheargumentusedtoderive(2),wehave(here∑ n
i=1
a
i
playstheroleofkin(2)):
(cid:0) B (cid:1)(cid:0) N B (cid:1)
P[I =a ,...,I =a]= ∑n i=1 ai n ∑n i=1 ai • 1 . (4)
1 1 n n (cid:0)N(cid:1) (cid:0) n (cid:1)
n ∑n

$i=1$
a_i

Animportantfeatureof(4)isthattheright-handsidedependson$(a_1,\ldots,a_n)$onlythroughtheirsum$\sum_{i=1}^{n} a_i$.

Inparticular,thejointdistributionisinvariantunderpermutationsoftheindices. Thatis,foranypermutation π of$\{1,\ldots,n\}$,

$$P[I_{\pi(1)}=a_1,\ldots,I_{\pi(n)}=a_n]=P[I_1=a_1,\ldots,I_n=a_n] \quad (5)$$

foralla$_1,\ldots,a_n\in\{0,1\}$.

Randomvariablessatisfying(5)arecalledexchangeable.

定义 16.5 (Exchangeability). A collection of random variables $X_1,\ldots,X_n$ with common 值域 A is saidtobeexchangeableif$(X_{\pi(1)},\ldots,X_{\pi(n)})$hasthesamejointdistributionas$(X_1,\ldots,X_n)$foreverypermutationπ of$\{1,\ldots,n\}$.

因此,$I_1,\ldots,I_n$ areexchangeablerandomvariables.

Wenowrecordseveralconsequencesofexchangeability.

1. Identicalmarginals. TherandomvariablesX$_1,\ldots,X_n$ haveidenticalmarginaldistributions:

$$P[X_1=a]=P[X_2=a]=\cdots=P[X_n=a] \quad (6)$$

foralla$\in A$.

Thisfollowsbecause,foranyiandanya$\in A$,

$$\begin{aligned}P[X_i=a]&=\sum_{b_1,\ldots,b_{i-1},b_{i+1},\ldots,b_n\in A} P[X_1=b_1,\ldots,X_{i-1}=b_{i-1},X_i=a,X_{i+1}=b_{i+1},\ldots,X_n=b_n]\\&=\sum_{b_1,\ldots,b_{i-1},b_{i+1},\ldots,b_n\in A} P[X_1=a,\ldots,X_{i-1}=b_{i-1},X_i=b_1,X_{i+1}=b_{i+1},\ldots,X_n=b_n]\\&=P[X_1=a],\end{aligned}$$

whereweusedexchangeabilityinthesecondstep.

AlthoughtheX$_i$ haveidenticaldistributions,theyaregenerallynotindependent. Infact,i.i.d.蕴含exchangeability,buttheconverseneednothold.

2. Identicalpairwisedistributions. Foranydistincti,j,

$$P[X_i=a,X_j=b]=P[X_1=a,X_2=b] \quad (7)$$

foralla,b$\in A$.

Theprooffollowsbysummingovertheremainingvariablesandapplyingexchangeability.

3. Identical k-tuple distributions. More generally, 对于 any distinct indices $i_1,\ldots,i_k$, 其中 $k<n$, the

1 k
jointdistributionof(X ,...,X)isthesameasthatof(X ,...,X).
Theproofissimilartotheprevious
i 1 ik 1 k
cases.

4 期望

The 分布 of a r.v. contains all the information about the r.v.
In most applications, 然而, the
完全的 分布 of a r.v. is very hard to calculate. 对于 例子, 考虑 the 作业
permutation
例子 with n=20. In principle, we would have to enumerate
20!≈2.4×1018 sample points, 计算
the value of X at each one, 和 count the number of points at
其中 X takes on each of its possible values
(though in practice we could streamline this calculation a
bit)! 此外, even when we can 计算 the
completedistributionofar.v.,itisoftennotveryinformative.
Forthesereasons,
weseektosummarizethedistributionintoamorecompact,
convenientformthatisalso
easiertocompute.
Themostwidelyusedsuchformistheexpectation(ormeanoraverage)ofther.v.
Definition16.6(期望). TheexpectationofadiscreterandomvariableX
isdefinedas
E[X]= ∑ a×P[X =a], (8)
a∈A
wherethesumisoverallpossiblevaluestakenbyther.v.

TechnicalNote.
Expectationiswelldefinedprovidedthatthesumontherighthandsideof(8)isabsolutely
convergent, i.e., ∑ a∈A |a|×P[X =a]<∞. 存在 离散的 random
variables 对于 其中 expectations do
非 exist, such as the r.v. X with 分布 P[X =i]= 6 对所有 正的
integers i. (The reason 对于 the
π2i2
factor 6 hereistomaketheprobabilitiessumto1,因为∑∞ 1 = π2 .)
π2 i=1i2 6

Foroursimplerpermutationexamplewithonly3students,theexpectationis
(cid:18) (cid:19) (cid:18) (cid:19) (cid:18) (cid:19)
1 1 1 1 1
E[X]= 0× + 1× + 3× =0+ + =1.
3 2 6 2 2

Thatis,theexpectednumberoffixedpointsinapermutationofthreeitemsisexactly1.
The 期望 can be seen in some sense as a “typical” value of the
r.v. (though 笔记 that E[X] may 非
actuallybeavaluethatX cantake).
Thequestionofhowtypicaltheexpectationisforagivenr.v.isavery
importantonethatweshallreturntoinalaterlecture.
Here is a physical interpretation of the 期望 of a random 变量:
imagine carving out a wooden
cutoutfigureoftheprobabilitydistributionasinFigure4.
Thentheexpectedvalueofthedistributionisthe
balancepoint(directlybelowthecenterofgravity)ofthisobject.
Figure4:
Physicalinterpretationofexpectedvalueasthebalancepoint.

4.1 Examples

1. Single die. Throw a fair die once 和 令 X be the number
that comes up. 那么 X takes on values

1,2,...,6eachwithprobability 1,所以
6
1 21 7
E[X]= (1+2+3+4+5+6)= = .
6 6 2
NotethatX neveractuallytakesonitsexpectedvalue 7.
2
2. Twodice. ThrowtwofairdiceandletX bethesumoftheirscores.
ThenthedistributionofX is
a 2 3 4 5 6 7 8 9 10 11 12
P[X =a] 1 1 1 1 5 1 5 1 1 1 1
36 18 12 9 36 6 36 9 12 18 36
Theexpectationistherefore
(cid:18) (cid:19) (cid:18) (cid:19) (cid:18) (cid:19)
(cid:18) (cid:19)
1 1 1 1
E[X]= 2× + 3× + 4× +・・・+ 12× =7.
36 18 12 36

3. Roulette.
Aroulettewheelisspun(recallthataroulettewheelhas38slots:
thenumbers1,2,...,36,
halfofwhicharereddandhalfblack,plus0and00,whicharegreen).
Youbet$1onBlack. Ifablack
number comes up, you receive your stake plus $1; otherwise
you lose your stake. 令 X be your net
winningsinonegame. ThenX
cantakeonthevalues+1and 1,andP[X=1]= 18,P[X= 1]= 20.
38 38
因此,
(cid:18) (cid:19) (cid:18) (cid:19)
18 20 1
E[X]= 1× +   1× =   ;
38 38 19
i.e., you expect to lose about a nickel per game. Notice how
the zeros tip the balance in favor of the
casino!
4.2 Linearity of 期望
Sofar,wehavecomputedexpectationsbybruteforce:
i.e.,wehavewrittendownthewholedistributionand
那么 added up the contributions 对所有 possible values of the
r.v. The 实数 power of expectations is that in
many 实数-life examples they can be computed much more easily
using a simple shortcut. The shortcut is
thefollowing:
Theorem16.1. ForanytworandomvariablesX andY
onthesameprobabilityspace,wehave
E[X+Y]=E[X]+E[Y].
Also,foranyconstantc,wehave
E[cX]=cE[X].
证明.
Wefirstrewritethedefinitionofexpectationinamoreconvenientform.
RecallfromDefinition16.6
that
E[X]= ∑ a×P[X =a].
a∈A
考虑 a particular term a×P[X =a] in the above sum. Notice that
P[X =a], by 定义, is the sum
of P[ω] over those sample points ω 对于 其中 X(ω)=a. 此外, we know
that every sample point

ω ∈ isinexactlyoneoftheseeventsX =a.
Thismeanswecanwriteouttheabovedefinitioninamore
long-windedformas
E[X]= ∑ X(ω)×P[ω]. (9)
ω∈
This 等价 定义 of 期望 will make the present 证明 much easier
(though it is usually less
convenientforactualcalculations). Applying(9)toE[X+Y]gives:
E[X+Y]= ∑ (X+Y)(ω)×P[ω]
ω∈
= ∑ (X(ω)+Y(ω))×P[ω]
ω∈
= ∑ (X(ω)×P[ω])+ ∑ (Y(ω)×P[ω])
ω∈ ω∈
=E[X]+E[Y]
Inthelaststep,weused(9)twice.
This completes the 证明 of the first equality. The 证明 of the
second equality is much simpler 和 is left
asanexercise.

定理 16.1 is very powerful: it says that the 期望 of a sum of
r.v.’s is the sum of their expecta-
tions,withnoassumptionsaboutther.v.’s.
WecanuseTheorem16.1toconcludethingslikeE[3X 5Y]=
3E[X] 5E[Y],regardlessofwhetherornotX andY areindependent.
Thisimportantpropertyisknownas
linearityofexpectation.
Important caveat: 定理 16.1 does 非 say that E[XY] = E[X]E[Y],
或 that E(cid:2)1(cid:3) = 1 , etc. These
X E[X]
claims are 非 真 in general. It is only sums 和 differences 和
常数 multiples of random variables
thatbehavesonicely.
4.3 Applications of Linearity of 期望
NowletusseesomeexamplesofTheorem16.1inaction.
1. Twodiceagain.
Hereisamuchlesspainfulwayofcomputing E[X],whereX
isthesumofthescores
of the two dice. 笔记 that X =Y +Y , whereY is the score on
die i. We know from 例子 1 in
1 2 i
Section4.1thatE[Y]=E[Y]= 7.
所以,byTheorem16.1,wehaveE[X]=E[Y]+E[Y]=7.
1 2 2 1 2
2. More roulette. 假设 we play the roulette game mentioned in
节 4.1 n⩾1 times. 令 X
n
be our expected net winnings. 那么 X =Y +Y +・・・+Y , 其中 Y is
our net winnings in the
n 1 2 n i
ith play. We know from earlier that E[Y]= 1 对于 each i. 因此,
by 定理 16.1, E[X]=
i 19 n
E[Y]+E[Y]+・・・+E[Y]= n . Forn=1000, E[X]= 1000 ≈ 53,
soifyouplay1000games,
1 2 n 19 n 19
youexpecttoloseabout$53.
3. Fixedpointsofpermutations.
Letusreturntothehomeworkpermutationexamplewithanarbitrary

number n of students. 令 X 表示 the number of students who
receive their own 作业 after
n
shuffling (或 equivalently, the number of fixed points). To
take advantage of 定理 16.1, we need
towriteX asasumofsimplerr.v.’s. SinceX
countsthenumberoftimessomethinghappens,wecan
n n
writeitasasumusingthefollowingusefultrick:
(cid:40)
1, ifstudentigetstheirownhomework,
X =I +I +・・・+I , whereI = (10)
n 1 2 n i
0, otherwise.
[Youshouldthinkaboutthisequationforamoment.
RememberthatalltheI’sarerandomvariables.
i
What does an 方程 involving random variables mean? What we
mean is that, at every sample
pointω,wehaveX (ω)=I (ω)+I (ω)+・・・+I (ω). Whyisthistrue?]
n 1 2 n
A Bernoulli random 变量 such as I is called an indicator
random 变量 of the corresponding
i
事件(inthiscase,theeventthatstudentigetstheirownhomework).
Forindicatorr.v.’s,theexpecta-
tionisparticularlyeasytocalculate. Specifically,
E[I]=(0×P[I =0])+(1×P[I =1])=P[I =1].
i i i i
Inourcase,wehave
1
P[I =1]=P[studentigetstheirownhomework]= .
i
n
WecannowapplyTheorem16.1to(10),yielding
1
E[X]=E[I]+E[I]+・・・+E[I]=n× =1.
n 1 2 n
n
所以, we see that the expected number of students who get
their own homeworks in a class of size n
is 1. 即, the expected number of fixed points in a random
permutation of n items is always 1,
regardlessofn!

4. Cointosses. Tossafaircoinn≥1times. Letther.v.X
bethenumberofheadsobserved. Asinthe
n
previousexample,totakeadvantageofTheorem16.1wewrite
X =I +I +・・・+I ,
n 1 2 n
whereI istheindicatorr.v.oftheeventthattheithtossisH.
Sincethecoinisfair,wehave
i
1
E[I]=P[I =1]=P[ithtossisH]= .
i i
2
UsingTheorem16.1,wethereforeget

n 1 n
E[X]= Σ = .
n
2 2
i=1
Moregenerally,inntossesofabiasedcointhatcomesupH
withprobability p,E[X]=np. (Check
n
this!) 所以 the 期望 of a 二项的 r.v. X Bin(n,p) is equal to np.
笔记 that it would have
been much messier (though possible) to reach the same 结论 by
computing this directly from

thedefinitionofexpectationin(8)andthedistributionofabinomialr.v.in(1).
5. Sampling without replacement. 考虑 sampling without
replacement n≤N balls from an urn
containingBblackballsandN Bwhiteballs. LetY
denotethenumberofblackballsinthesample.
WeknowthatY followstheHypergeometric(N,B,n)分布.
然而,computingE[Y]directly
from the 分布 (see (3)) can be cumbersome. A much simpler
approach is to use indicator
randomvariables. Write
Y =I +・・・+I ,
1 n
其中 I is the indicator that the ith draw is black. 虽然 the
random variables I ,...,I are 非
i 1 n
独立的,linearityofexpectationstillapplies:
E[Y]=E[I]+・・・+E[I].
1 n
此外,byexchangeability,theI haveidenticaldistributions,所以
i
E[Y]=nE[I].
1
Finally,sincetheprobabilitythatthefirstdrawisblackisP[I =1]=
B,weconcludethat
1 N
B
E[Y]=n .
N
6. Ballsandbins. Throwmballsintonbins. Letther.v.X
bethenumberofballsthatlandinthefirst
bin. 那么 X behaves exactly like the number of heads in m
tosses of a biased coin with P[H]= 1
n
(why?). 所以, from the previous 例子, we get E[X]= m. In the
special case m=n, the expected
n
numberofballsinanybinis1.
Ifwewantedtocomputethisdirectlyfromthedistributionof X, we
wouldgetintoamessycalculationinvolvingbinomialcoefficients.
Here is another 例子 on the same 样本空间. 令 the r.v. Y be the
number of empty bins.
n
The 分布 of Y is horrible to contemplate: to get a feel 对于
this, you might like to write it
n

down 对于 m=n=3 (i.e., 3 balls, 3 bins). 然而, computing the 期望
E[Y] is easy using
n
Theorem16.1. Asinthepreviousexamples,wewrite
Y =I +I +• • • +I , (11)
n 1 2 n

whereI istheindicatorr.v.oftheevent“biniisempty”.
TheexpectationofI iseasytofind:
i i
(cid:18) (cid:19)m
1
E[I]=P[I =1]=P[biniisempty]= 1 ,
i i
n
asdiscussedearlier.
ApplyingTheorem16.1to(11),wethereforeobtain
n (cid:18) 1 (cid:19)m
E[Y]= ∑E[I]=n 1 ,
n i
n
i=1
a simple formula, quite easily derived. 令 us see how it
behaves in the special case m=n (same
number of balls as bins). In this case we get E[Y]=n
(cid:0) 1 1(cid:1)n . Now the quantity (cid:0) 1 1(cid:1)n
can be
n n n
approximated(forlargeenoughvaluesofn)bythenumber 1.2
Soweseethat,forlargen,
e
n
E[Y]≈ ≈0.368n.
n
e
The bottom line is that, 如果 we throw (say) 1000 balls into
1000 bins, the expected number of empty
binsisabout368.
2Moregenerally,itisastandardfactthatforanyconstantc,
(1+c)n→ec asn→∞.
n
Wejustusedthisfactinthespecialcasec= 1.
Theapproximationisactuallyverygoodevenforquitesmallvaluesofn.
(Tryit
yourself!)
E.g.,forn=20wealreadyget(1 1)n≈0.358,whichisveryclosetoto 1
≈0.368. Theapproximationgetsbetterand
n e
betterforlargern.